

Machine Learning Exercise Sheet 10

Dimensionality Reduction & Matrix Factorization, Part 1

In-class Exercises

Problem 1: In this exercise, we use proof by induction to show that the linear projection onto an M -dimensional subspace that maximizes the variance of the projected data is defined by the M eigenvectors of the data covariance matrix $\mathbf{S}$, given by

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T \quad \bar{\mathbf{x}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$$

corresponding to the M largest eigenvalues. In Section 12.1 in Bishop this result was proven for the case of $M = 1$. Now suppose the result holds for some general value of M and show that it consequently holds for dimensionality $M + 1$.

Suppose that the result holds for projection spaces of dimensionality M . The $M + 1$ dimensional principal subspace will be defined by the M principal eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_M$ together with an additional direction vector $\mathbf{u}$ whose value we wish to determine.

Following the argument given in section 12.1.1 for $\mathbf{u}_1$, we see that the variance of the data projected onto the subspace spanned by $\{\mathbf{u}_1, \dots, \mathbf{u}_M, \mathbf{u}\}$ is given by $\mathbf{u}^T \mathbf{S} \mathbf{u} + \sum_{i=1}^M \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i$. Since $\{\mathbf{u}_1, \dots, \mathbf{u}_M\}$ are already chosen and fixed, our goal is to maximize $\mathbf{u}^T \mathbf{S} \mathbf{u}$.

We must constrain $\mathbf{u}$ such that it cannot be linearly related to $\mathbf{u}_1, \dots, \mathbf{u}_M$ (otherwise it will lie in the M -dimensional projection space instead of defining an $M + 1$ independent direction). For this, we introduce M constraints that require $\mathbf{u}$ to be orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_M$. These constraints are enforced using Lagrange multipliers $\eta_1, \dots, \eta_M$.

Finally, we introduce a constraint with Lagrange multiplier λ that ensures that $\mathbf{u}^T \mathbf{u} = 1$.

Putting everything together, we obtain the Lagrangian

$$L(\mathbf{u}, \lambda, \eta_1, \dots, \eta_M) = \mathbf{u}^T \mathbf{S} \mathbf{u} + \lambda(1 - \mathbf{u}^T \mathbf{u}) + \sum_{i=1}^M \eta_i \mathbf{u}_i^T \mathbf{u}$$

The solution to a constrained optimization problem is found at the points $(\mathbf{u}, \lambda, \eta_1, \dots, \eta_M)$, where the gradient of the Lagrangian w.r.t. $\mathbf{u}$ is zero and all other constraints are fulfilled, that is

$$\begin{cases} \nabla_{\mathbf{u}} L(\mathbf{u}, \lambda, \eta_1, \dots, \eta_M) = 2\mathbf{S}\mathbf{u} - 2\lambda\mathbf{u} + \sum_{i=1}^M \eta_i \mathbf{u}_i = \mathbf{0} \\ \mathbf{u}^T \mathbf{u} = 1 \\ \mathbf{u}^T \mathbf{u}_i = 0 \text{ for } i = 1, \dots, M \end{cases}$$

By left-multiplying the first equality with $\mathbf{u}_j^T$, we can conclude that it must hold $\eta_j = 0$ for $j = 1, \dots, M$. This means we can simplify the problem to

$$\begin{cases} \mathbf{S}\mathbf{u} = \lambda\mathbf{u} \\ \mathbf{u}^T\mathbf{u} = 1 \\ \mathbf{u}^T\mathbf{u}_i = 0 \text{ for } i = 1, \dots, M \end{cases}$$

This set of equalities is fulfilled for any eigenvalue-eigvector pair $(\lambda_i, \mathbf{u}_i)$ of $\mathbf{S}$, except the first M eigenvalues/eigenvectors, since those are not orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_M$.

Now we just need to select one of feasible solutions $\{\mathbf{u}_{M+1}, \dots, \mathbf{u}_D\}$ that maximizes the variance:

$$\arg \max_{i \in \{M+1, \dots, D\}} \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i$$

This happens to be the $M + 1$'s eigenvector $\mathbf{u}_{M+1}$ that corresponds to variance λ_{M+1} , since we assumed w.l.o.g. that the eigenvalues are sorted in decreasing order.

To summarize, the variance in the projected space is maximized by choosing $\mathbf{u}$ to be the eigenvector having the largest eigenvalue amongst those not previously selected. Thus the result holds also for projection spaces of dimensionality $M + 1$, which completes the inductive step. Since we have already shown this result explicitly for $M = 1$ it follows that the result must hold for any $M \leq D$.

Problem 2: Proof that minimizing the error is equivalent to maximizing the variance.

See Bishop chapter 12.1.2.